

JavaScript ES6+ Coding Practice Sheet (25 Questions)

1. Create a variable using `const` and store your college name. Print it.
2. Create a variable using `let` and update its value three times.
3. Create a block using `{}` and show that a variable declared with `let` cannot be accessed outside it.
4. Write a program to swap two numbers using array destructuring.
5. Create an arrow function that returns the cube of a number.
6. Create an arrow function to check whether a number is even or odd.
7. Create an arrow function that finds the maximum of three numbers.
8. Given an array:

JavaScript

```
const nums = [10, 20, 30, 40, 50];
```

Use destructuring to get first, second and remaining values.

9. Create a function that receives a user's name and age and returns: "Hello Ram, you are 25 years old." using template literals.
10. Generate the following URL dynamically: `/api/users/101` using template literals.
11. Create an object and destructure all properties:

JavaScript

```
const student = {  
  name: "Ram",  
  age: 25,  
  course: "MERN"  
};
```

12. Rename `course` to `technology` while destructuring.
13. Create a function with a default parameter: `createUser(name, role="Student")`
14. Create a calculator function where the operation defaults to `"add"`.
15. Create a function that receives unlimited numbers and returns their sum using the Rest Operator.
16. Create a function that receives unlimited numbers and returns the largest number.
17. Merge two arrays using the Spread Operator:

JavaScript

```
const frontend = ["HTML", "CSS"];
const backend = ["Node", "Express"];
```

18. Copy an array using the Spread Operator and add one extra value.
19. Copy an object and update only the `email` property.
20. Create a product object and create a new object with an added `discount` field using the Spread Operator.
21. Use `map()` and an arrow function to return only names from the given array:

JavaScript

```
const users = [
  { name: "Ram", age: 25 },
  { name: "Mohan", age: 30 },
  { name: "Amit", age: 22 }
];
```

22. Use `filter()` to get users whose age is greater than 24.
23. Use `reduce()` to calculate the total age of all users.
24. Create a function `createInvoice(customerName, amount)` and return a formatted string using template literals.
25. **25. Mini Challenge**

Given the student object:

JavaScript

```
const student = {
  name: "Ram",
  marks: [80, 90, 70, 85]
};
```

Using Destructuring, Arrow Functions, Rest/Spread, and Template Literals, calculate the total marks and average, then print:

None

```
Student: Ram
Total: 325
```

